

ASIM ALTAF

Backend Engineer - Go · Telecom, Signalling & Distributed Systems at Scale

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PROFILE

Backend engineer building production telecom and distributed systems in Go, C, and C++. At Safarifone, I work across 5G core UDM/UDR, subscriber identity resolution (SUPI/SUCI), eSIM provisioning, SS7/HLR signalling, and subscriber-platform services for a multi-million subscriber base. Before that, I built VoIP/SIP infrastructure in Go handling 10,000+ calls/second. I care about correctness, latency, observability, and maintainable systems that keep working under real production load.

EXPERIENCE

Backend / Systems Engineer · Safarifone (mobile network operator)

Nov 2025 - Present · Islamabad

Work across four production teams at multi-million subscriber scale: subscriber authentication (5G core), eSIM provisioning, SS7 signalling, and the subscriber platform.

Own subscriber authentication for our 3GPP UDM/UDR (the 5G-generation successor to the HSS): implemented SUPI/SUCI resolution and authentication-vector generation to TS 29.503/29.505/23.502 in C with libuv.

Replaced a thread-heavy design in that authentication path with an async event loop, cutting p95/p99 latency measurably. Profiled a 50K+ line C codebase with Valgrind to ship the rewrite without regressions.

Architected our eSIM provisioning service in Go against a live SM-DP+ (GSMA SGP.02, ES2+): built the mTLS client-authentication layer, callback handling, and the full profile-issuance state machine end to end, order to installed profile.

Built the HLR/MAP adapter (SendRoutingInfo, UpdateLocation, InsertSubscriberData) and a TCAP session/transaction manager for USSD over SIGTRAN, in Go - including a custom zero-allocation binary framing protocol and the SS7-to-JSON translation layer other teams build on.

Day to day: Go, C/C++, gRPC/Protobuf, Kubernetes, Docker, Temporal.io, PostgreSQL, Oracle, Redis.

Golang Developer (part-time remote contract) · ContactVA - US enterprise clients

Sep 2024 - Jul 2026 · Remote

Built VoIP/SIP infrastructure in Go: a UDP signalling server sustaining 10,000+ calls/second, with SIP signalling, RTP relay and OpenSIPS call routing - production experience running network-facing Go services under continuous real-world load.

Cut API latency from ~1,000ms to ~300ms by reworking OpenSIPS flows and the PostgreSQL/Redis data layer.

Ran alongside the Safarifone role from late 2025, with the client's agreement, until the contract concluded in July 2026.

Backend Developer · ZIXEL

Sep 2023 - Jun 2024 · Islamabad

First role out of university. Built and maintained Go microservices across five products.

Added caching and autoscaling that cut response times by roughly 20%; mentored two junior backend developers.

TECHNICAL SCOPE

Authentication & security: cryptographic authentication-vector generation (SUPI/SUCI) to 3GPP spec, mTLS client authentication, session/transaction state management (TCAP), hash-based identity resolution.

Languages & systems: Go (production, multi-year), C, C++ (libuv), concurrency & event-loop design, gRPC/Protobuf.

Data & infra: PostgreSQL, Oracle, Redis, Kubernetes, Docker, Temporal.io.

Signalling & core network: SS7 (MAP, TCAP, USSD, SIGTRAN), HLR/HSS, 5G SBA (UDM/UDR), SIP/RTP.

Web fundamentals: HTTP, JSON APIs, service-to-service authentication, and mTLS-secured integrations.

EDUCATION

BS Computer Science - National University of Computer & Emerging Sciences (FAST-NUCES)

2019 - 2023

Final Year Project selected for a funded incubation program, supporting further development and commercialization.

CERTIFICATIONS

5G Network Fundamentals · Go - Complete Course · AWS Certified Cloud Practitioner (in progress)